Acoustic Data-based Characterization of Incipient Boiling for Space Applications: A Progress Report

Hailey Ernest
Statistics Dept.
Cal Poly, SLO
hernest@calpoly.edu

Elliot Kunz
Mathematics Dept.
Cal Poly, SLO
edkunz@calpoly.edu

Cameron Hafer
Statistics Dept.
Cal Poly, SLO
clhafer@calpoly.edu

Colin Hassett
Statistics Dept.
Cal Poly, SLO
cmhasset@calpoly.edu

Abstract

Detecting incipient boiling caused by localized heat leaks is important for future space-based cryogenic fuel systems. This project explores whether acoustic accelerometer data could be used to distinguish meaningful boiling behaviors through unsupervised machine learning. Building on prior work, our team reviewed earlier models, engineered new signal features, tested new dimensionality reduction and clustering methods, and improved visualization of cluster behavior and separation. The goal was to produce results that were scientifically useful and still understandable to non-technical stakeholders. Our analysis showed clearer separation between boiling behaviors, specifically between rhythmic and non-rhythmic patterns. After integrating new accelerometer-based features to the current raw amplitude-based features, our clustering produced more interpretable groupings and a RAND index of about 73%, indicating moderately strong clustering stability across parameter changes. These results show that acoustic data can help identify meaningful boiling regimes and support future interpretable machine learning tools for boiling diagnostics. The analysis also revealed substructure within the rhythmic regime, suggesting additional physically meaningful boiling subtypes.

1. Introduction

Detecting early-stage boiling is important for maintaining stability in cryogenic fuel systems, especially in space environments where heat transfer behaves differently than on Earth. We focused on building a clustering framework to group similar boiling signals and better understand differences between boiling behaviors. By grouping similar signals together, we could begin to understand what features separate different types of boiling behavior and interpret what each cluster represents physically. This allows us to move from raw signal data toward meaningful physical insight. We engineered 32 features from the signal data and used UMAP and HDBSCAN to cluster the data.

2. Project Goals

Our goal for this project was to produce meaningful clusters of acoustic boiling regimes so we could find interpretable differences between distinguishable groups. The previous group produced clusters and visualizations using Principal Component Analysis (PCA) for dimensionality reduction and K-means clustering for clustering the regimes. However, these approaches assume a simple structure and might not capture more complex or irregular relationships in the data. We wanted to explore this by using a different dimensionality reduction and clustering algorithm to see if we could make a clearer separation between groups. Specifically, we explored UMAP for non-linear dimensionality reduction and HDBSCAN for density-based clustering. These methods could work better with complex, real-world data and relationships. We also wanted to incorporate more feature engineering to extract more info and produce better clusters. The focus was designing features that capture signal structure (such as frequency and variability) and depend less on amplitude, which can group noise into clustering.

3. Background

3.1 - Experimental Setup

The experimental design of this work follows exactly from experimental setup in M. Khasin, J. Rogers, and V. Osipov, “Acoustic Data-based Characterization of Incipient
Boiling for Space Applications. Below, in figure 1, is a diagram of their experiment.

The system includes a cryogenic tank with accelerometers placed on its surface. A controlled heat source induces boiling, and the resulting acoustic signals are recorded. These signals capture the physical behavior of bubble formation and collapse during boiling.

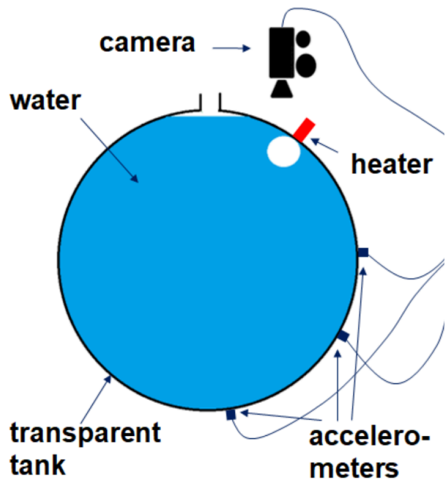

Figure 1.—Experimental setup: (M. Khasin, J. Rogers, and V. Osipov, “Acoustic Data-based Characterization of Incipient Boiling for Space Applications, ”)

3.2 - Data Format and Visualizations

The data that we obtained has 446 runs with lengths between 0.1 seconds and 30 seconds. Each run represents a recording of boiling activity, where signal variations correspond to changes in bubble dynamics over time. Both accelerometers transmitted data, so we visualize both of their data using the same visual and make sure that both of the signals are seen. Both accelerometers transmit data, so we visualize their signals together. Using both accelerometers lets us capture differences in boiling signals, which could provide more information for distinguishing between boiling behaviors.

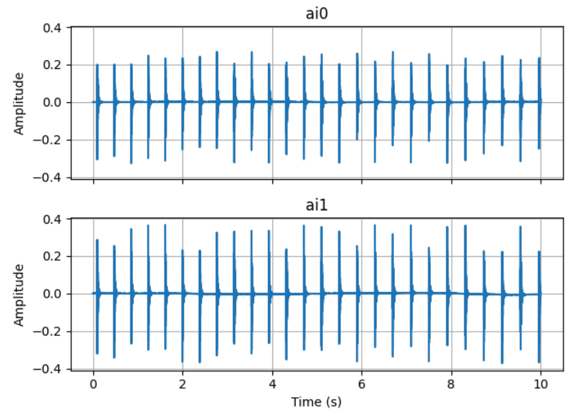

Figure 2: Raw signal from a representative boiling experiment with the data gathered from the first accelerator above the signal gathered by the second accelerator. (Run: MATLAB 3-26 PM Wed, Sep 18, 2024 Run12)

4. Methods

4.1 Review of Previous Work

After reviewing the previous work done in Gupta, K., et al., “Interpretable Machine Learning for Acoustic Classification of Incipient Boiling Regimes," we discovered a large error in the previous work's assumptions. Previous work believed that there were no differences between the signals, which led them to only use the signal transmitted from the first accelerometer for each run. However, as you can see in Figure 3 below, after exploring the signals ourselves, we discovered that there were noticeable differences.

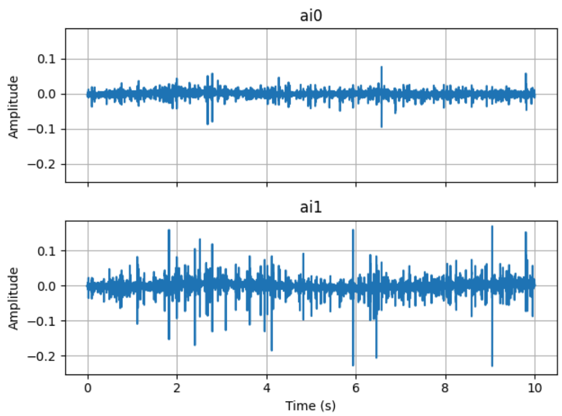

Figure 3: Representative sample showing the large differences in signal between accelerometer 1 and accelerometer 2 in both magnitude and time of peak amplitudes. (Run: MATLAB 2-25 PM Tue, Jun 4, 2024 Run3)

In reviewing previous work, we also discovered that there were multiple features like max amplitude, which can cause bad clustering due to the variability in amplitude that does not have any correlation to rhythmic vs. random boiling.

4.2 Feature Engineering

To improve representation of boiling behavior, we engineered 32 features from both time and frequency domains, whereas the previous group used only 15. These features were designed to capture the structure and dynamics of the signal and not rely solely on magnitude. We implemented two pairwise features: the number of peak magnitudes with high differences and the correlation between peak magnitudes. This way, differences in signal data between the two accelerometers could be used for clustering. With boiling runs not being the same in length, we removed linear effects of runtime from each feature via regression. We removed features that relied directly on raw magnitude, like max peak, but kept features that measured the variability of magnitude. We kept the previous group's spectral transformation where they used a high-pass Butterworth filter, then used FFT, and then used Welch's method to compute spectral features. We now use it to compute spectral flux, mean spectral entropy time, and variance of spectral centroid time.

Because runs vary in length, we checked whether any features were correlated with duration of runs, which could lead to clustering that depends on time instead of signal behavior. We calculated correlations between each feature and the length of a run and found that multiple features are highly correlated with time (see Appendix for a full list of features). To remedy this problem, we used detrending algorithms. The detrending process removes linear trends from each selected feature to ensure that clustering is not biased by systematic changes over time. A linear regression model is fit between each feature and a time variable, and the resulting trend is subtracted from the original feature values.

4.3 Dimensionality Reduction and Clustering

To analyze the feature space, we tested multiple dimensionality reduction methods, including PCA, diffusion maps, and UMAP. Although PCA was useful as a baseline, UMAP produced clearer visual separation between groups and was more effective for interpreting cluster structure. For clustering, we compared K-Means, HDBSCAN and spectral clustering. K-Means had been used by the previous group, but HDBSCAN proved more useful because it could identify clusters of varying density and handle outliers in a more fitting way. HDBSCAN also proved more useful in detecting total cluster amounts greater than two, which has proved extremely crucial in detecting boiling regimes. Some boiling runs did not fit neatly into one dense group because outlier identification helped distinguish structured behavior from noise-like behavior. Our final workflow focused on UMAP for visualization and HDBSCAN for clustering, with repeated testing under different parameter tunings to optimize stability and separation.

It is important to note that unsupervised clustering without a dimension reduction algorithm was tested. With all three clustering techniques, the result often failed to cleanly distinguish between boiling regimes. Noise and outliers are highly prominent in our feature set, meaning some kind of cleaning or reduction of noise must be present to effectively apply a clustering technique.

4.4 Cluster Validation

To assess the consistency of the clustering structure, we used the RAND index as a validation metric. The RAND index allowed us to compare clustering consistency across parameter changes and see whether runs continued to group in stable ways. Our final analysis got a RAND index of approximately 73%, indicating relatively strong consistency for this unsupervised problem. This result showed that the clustering patterns were not just random or too sensitive to parameter tunings. It could be recognizing that some disagreement may come from errors in the human-labeled data.

5. Results

The final clustering results showed a clear separation between rhythmic and non-rhythmic boiling behaviors in the UMAP embedding space. Rhythmic signals showed structured patterns that included a few subclusters, and non-rhythmic signals showed compact patterns. To continue investigating this structure, we performed subclustering within the rhythmic group, which allowed us to identify more specific patterns within rhythmic behavior that were not clear in the initial clustering. This suggests that rhythmic boiling contains underlying structure that the model is able to separate into subtypes.

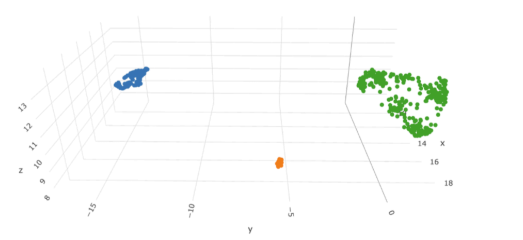

Figure 4 : 3D UMAP and HDBScan Clustering

These subclusters show that rhythmic boiling is not a single behavior and it has multiple distinct patterns with different timing and intensity characteristics. Visualization improvements, including consistent axis scaling and zoomed-in cluster views, made these patterns easier to interpret and compare across runs.

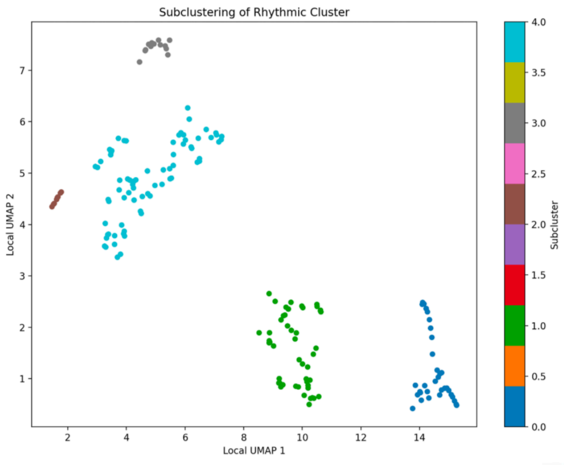

Figure 5 : Subclustering of the Green Cluster From above

Below are representative runs from each cluster, each matching a labeled regime.

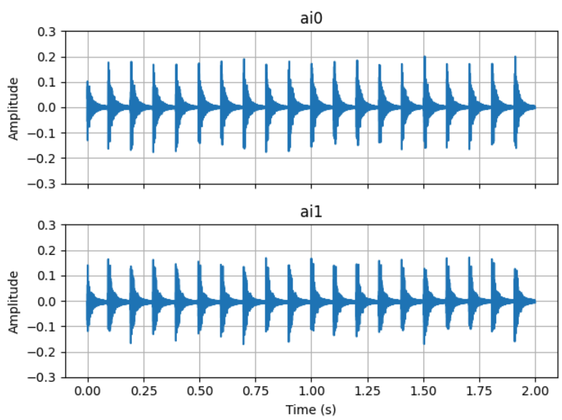

Figure 6 : Subcluster 0 of the Rhythmic Cluster - Single Rhythmic

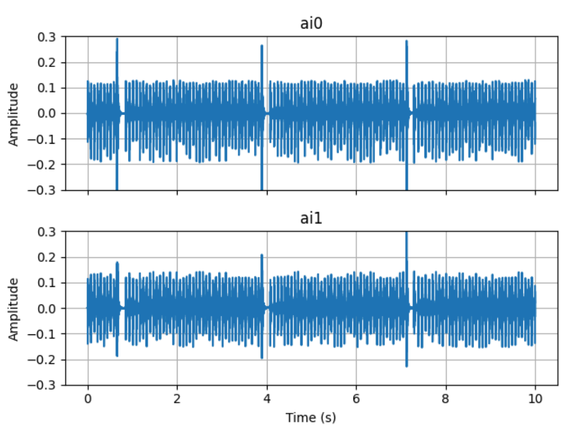

Figure 7 : Subcluster 1 of the Rhythmic Cluster - Rhythmic With Climax

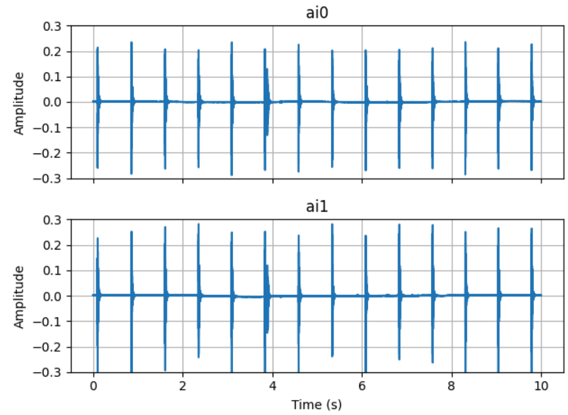

Figure 8 : Subcluster 2 of the Rhythmic Cluster - Single Rhythmic

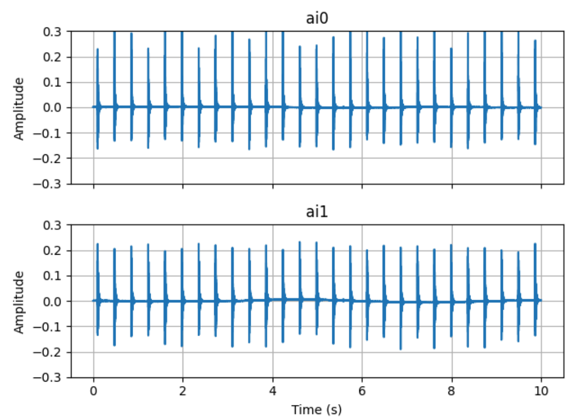

Figure 9 : Subcluster 3 of the Rhythmic Cluster - Single or Double Rhythmic

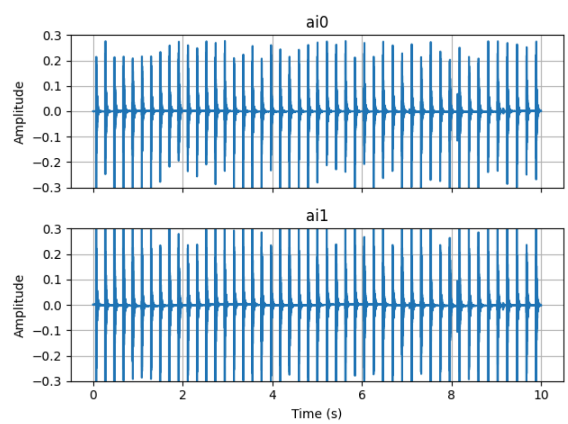

Figure 10 : Subcluster 4 of the Rhythmic Cluster - Double Rhythmic

6. Discussion and Future Work

We have demonstrated a clear distinction between rhythmic and non-rhythmic boiling using our methods. The subclustering results suggest that rhythmic boiling is not a single behavior and it consists of multiple distinct patterns with different timing and intensity characteristics. The research is now progressing toward subclustering within the rhythmic group identified through 3D UMAP. Although we have developed an initial model, we aim to further improve clustering performance, particularly by increasing the silhouette score.

We plan to use the results of this subclustering to compute the RAND Index across different UMAP and HDBSCAN parameter combinations. This will allow us to evaluate how consistently different parameter settings group similar runs, providing insight into the reproducibility of the method. We also intend to continue developing additional features to expand the feature space, with the goal of improving the overall accuracy and robustness of the results.

7. Limitations

This project had several limitations. The true number of boiling regimes is not known in advance, and no ground truth labels exist for the data. Because of this, clustering ability needs to be evaluated using comparison metrics and not accuracy. Additionally, some observed differences may be affected by experimental conditions that are not fully controllable in the feature set. Moreover, the interpretation of clusters also depends on human review of acoustic plots and is used when deciding whether a subgroup is physically meaningful.

By using a dimension reduction algorithm, we gained the ability to visualize our clusters in both 2 and 3 dimensions. This has proved essential for understanding the data and providing interpretable results, but it can lead to a loss of information and a risk of omitting subtle patterns. The differences between a dimension reduction algorithm and a noise reduction algorithm might exist even in a small way. An investigation into the differences between the two types of algorithms may prove useful results now that the general patterns of boiling regimes.

8. References

1. M. Khasin, J. Rogers, and V. Osipov, “Acoustic Data-based Characterization of Incipient
Boiling for Space Applications,” NASA Technical Memorandum NASA/TM–20250009668,
September 2025.

2. Gupta, K., et al., “Interpretable Machine Learning for Acoustic Classification of Incipient Boiling Regimes,” NASA Technical Memorandum NASA/TM–20250011359, Dec. 2025.

3. M. Khasin, “Acoustic and Visual Data for Incipient Boiling at Local Heat Leaks in a Water
Tank (v1.0),” NASA Intelligent Systems Division Datasets, NASA Ames Research Center, last updated Nov. 25, 2025.

Available:
https://www.nasa.gov/intelligent-systems-division/datasets/
Appendix

Features Subject to Detrending

The following features were selected for detrending during preprocessing:

1. RMS change points
2. High to low ratio
3. Spectral flatness
4. Peaks per second
5. Number of boilings
6. Mean time difference
7. Median time difference

Link to Features - Feature Explanations

Slides:
https://www.canva.com/design/DAHDlrj4ks4/jXvWJ4puMNnqIu_zqO_OMg/edit?utm_content=DAHDlrj4ks4&utm_campaign=designshare&utm_medium=link2&utm_source=sharebutton